

## WORKSHEET - NAPLAN Year 9

### Number and Algebra

#### Fractions, Decimals and Percentages (9)

Teacher: Joanne Rust

Exam Equivalent Time: 10 minutes (based on NAPLAN allocation of ~1.25 min/mark)



### Questions

1. Number and Algebra, NAP-85047

A return trip from Brodie's house to the beach is 5.78 kilometres.

How far does Brodie travel if he does this 14 times?

8.09 km    23.12 km    42.12 km    69.36 km    80.92 km

☐    ☐    ☐    ☐    ☐

2. Number and Algebra, NAP-61530

Which number is exactly halfway between  $1\frac{1}{3}$  and  $4\frac{2}{3}$ ?

$2\frac{1}{3}$     3     $3\frac{1}{3}$      $3\frac{2}{3}$

☐    ☐    ☐    ☐

3. Number and Algebra, NAP-02658

Which of these numbers is closest in value to  $\frac{338}{380}$ ?

0.6    0.7    0.8    0.9    1.1

☐    ☐    ☐    ☐    ☐

4. Number and Algebra, NAP-73219

Megan has  $\frac{1}{8}$  cup of sugar.

She needs  $1\frac{1}{2}$  cups of sugar for a cake recipe she is using.

How much more sugar does Megan need for the recipe?

$1\frac{1}{8}$  cups     $1\frac{1}{4}$  cups     $1\frac{3}{8}$  cups     $1\frac{5}{8}$  cups

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5. Number and Algebra, NAP-55671

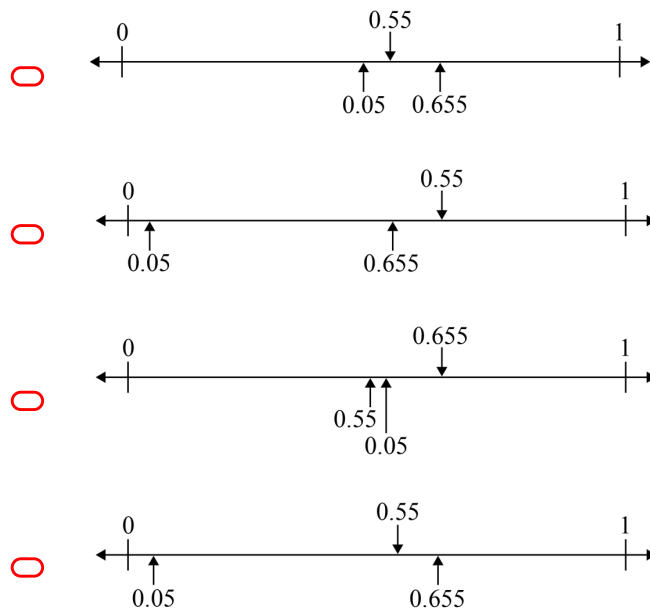
A high school has 1000 students enrolled.

37 of the 1000 students have the first name John.

Write this as a decimal.

6.  Number and Algebra, NAP-02684

Select the number line that correctly shows the positions of 0.05, 0.655 and 0.55.



7.  Number and Algebra, NAP-43816

The weight of a banana is 0.1246 kg.



What is the weight of the banana rounded to the nearest hundredth of a kilogram?

0.12 kg    0.13 kg    0.124 kg    0.125 kg

☐    ☐    ☐    ☐

8.  Number and Algebra, NAP-96803

A locksmith increases the price of replacing a key from \$12 to \$15.

What is the percentage increase her fee?

%

## Worked Solutions

1. Number and Algebra, NAP-85047

$$\begin{aligned}\text{Distance} &= 14 \times 5.78 \\ &= 80.92 \text{ km}\end{aligned}$$

2. Number and Algebra, NAP-61530

$$\begin{aligned}\text{Halfway} &= \left(1\frac{1}{3} + 4\frac{2}{3}\right) \div 2 \\ &= 6 \div 2 \\ &= 3\end{aligned}$$

3. Number and Algebra, NAP-02658

$$\begin{aligned}\frac{338}{380} &= 0.889\dots \\ \therefore 0.9 &\text{ is the closest.}\end{aligned}$$

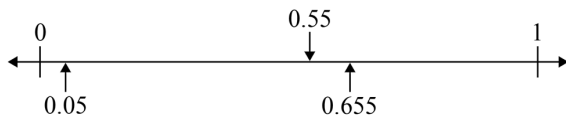
4. Number and Algebra, NAP-73219

$$\begin{aligned}\text{Sugar needed} &= 1\frac{1}{2} - \frac{1}{8} \\ &= 1\frac{3}{8} \text{ cups}\end{aligned}$$

5. Number and Algebra, NAP-55671

$$\frac{37}{1000} = 0.037$$

6. Number and Algebra, NAP-02684



## Worked Solutions

7. Number and Algebra, NAP-43816

Nearest hundredth - look at the first  
3 decimal places and round to 2.  
 $\therefore 0.12 \text{ kg}$

8. Number and Algebra, NAP-96803

$$\text{Price increase} = 15 - 12 = \$3$$

$$\begin{aligned}\text{Percentage increase} &= \frac{3}{12} \\ &= 0.25 \\ &= 25\%\end{aligned}$$